Business Process Engineering
Week # 10

FALL 2025

Plan–Do–Check–Act (PDCA)



- Framework for problem-solving, continuous improvement and change
- Widely recognized as the basis of continually improving the quality of processes, products and services
- Provides a simple and structured approach for solving quality-related problems
- Multiple iterations of the PDCA cycle may be necessary to solve the problem permanently and reach the ultimate goal
- Used to verify the feasibility of a problem whether it is an incremental or breakthrough improvement
- Often used when:
  - Intending to make minor changes to a process
  - Solution to the problem is known

PDCA Benefits

- Encourages the methodical way of problem solving and implementing solutions.
- Ensures that you plan, test and incorporate feedback before you start full-scale implementation. This brings you closer to your goals as knowledge is increased:
  - From solving problems.
  - From failures.
  - From the feedback received.
- Improves the critical thinking skills of your team.
- Helps to reach towards a more integrated system.

PLAN Step



- The team selects the problem to be solved (or the process to be improved).
- The problem and objectives are clearly identified.
- The current situation is analyzed.
- Solution alternatives are identified, selected and scheduled.

DO Step



- The solution is tested on a small scale basis.
- It involves collecting data for later analysis.
- It also involve measuring progress.
- It ensures the solution is appropriately tested and benefits are validated before committing to full implementation.

CHECK Step



- Involves analyzing the collected data and comparing the actual results against the planned objectives.
- Allows evaluating how well the solution worked.
- Allows discussing whether further improvements are possible.

- Concerned with identifying the unexpected issues, their causes, and gathering and summarizing the key learnings.

You may need to repeat the **Do** and **Check** a number of times until you get the optimum results

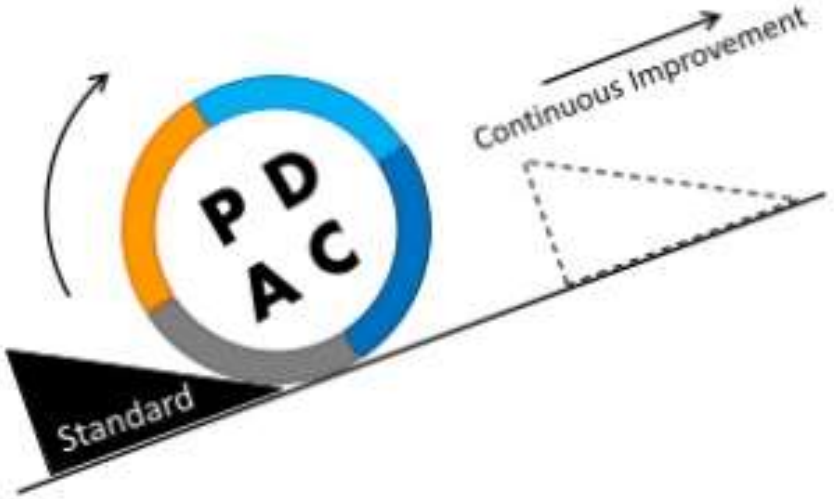
ACT Step



- Involves acting on the feedback and lessons learned and implementing the solution fully.
- **It is also concerned with:**
  - Standardizing.
  - Documenting.
  - Sustaining the improved process.
  - Integrating it into the organization's system.

Continuous Improvement Tool

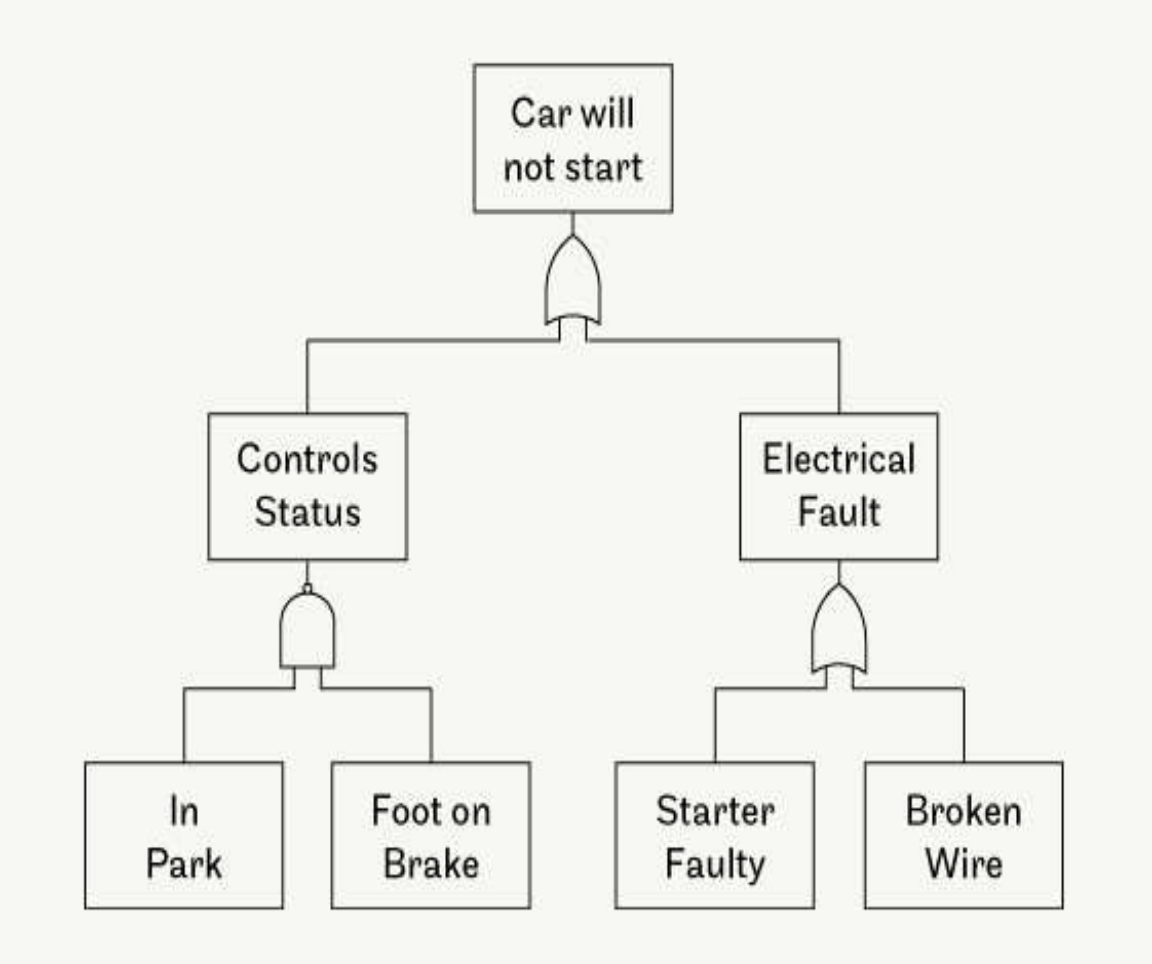
- You may identify the next target and start again at the plan phase.
- The PDCA cycle can be repeatedly applied in a process of continuous improvement where there is no end to it.
- Each cycle will bring you closer to your goals and will extend your knowledge further.



Tools used during PDCA Steps

Phase	Useful Tools
Plan	Gap analysis, fault tree analysis, waste analysis, brainstorming, process mapping, WBS, etc.
Do	Gantt charts, on-the-job training, conflict resolution, data collection methods, sampling, control charts, etc.
Check	Graphical analysis, cause and effect analysis, statistics, Pareto analysis, group decision-making technique, etc.
Act	Gantt charts, check sheets, control charting, control planning, standard work, conflict resolution, etc.

Fault Tree Analysis




12. Continuous Improvement
01. Select Project
11. Monitor Solution
02. Explain Reason
10. Implement Solution
03. Set Goals
09. Ensure Goals are Satisfied
04. Prepare Action Plan
08. Test Solutions
05. Gather The Data
07. Develop Solutions
06. Analyze The Facts
ACT
PLAN
DO
CHECK
Continuous Improvement Cycle

Toyota's A3 Method

- Toyota, with around 350,000 employees, is known for its continued commitment to improving operational performance with a Lean thinking tool called the A3 process
- A3 process is a problem solving tool Toyota developed to foster learning, collaboration, and personal growth in employees.
- The term "A3" is derived from the particular size of paper (11" X 17") used to outline ideas, plans, and goals throughout the A3 process.
- Toyota uses A3 reports for:
  - Problem Solving
  - Reporting Project Status
  - Proposing Policy Change

Why Use an A3 Process?

- Allows groups of people to actively collaborate on the purpose, goals, and strategy of a project
- Encourages in-depth problem solving throughout the process and adjusting as needed to ensure that the project most accurately meets its intended goal
- Helps an organization to foster effective collaboration and bringing out the best problem solving in teams
- Using the A3 process to foster collaboration can help organizations and teams invest their time, money, and momentum most effectively.

Steps of the A3 Process

- There are nine (well, ten) steps in the A3 process.
  - 0: Identify the problem
  - 1: Capture the current state of the situation
  - 2: Conduct a root cause analysis
  - 3: Devise countermeasures to address root causes
  - 4: Define your target state
  - 5: Develop a plan for implementation
  - 6: Develop a follow-up plan with predicted outcomes
  - 7: Get everyone on board
  - 8: Implement!
  - 9: Evaluate Results

0: Identify the problem

- The purpose of the A3 process is to solve problems or address needs
- First, undocumented/unwritten step is that you need to identify a problem or need

1: Capture the current state of the situation

- Observe the work processes and document your observations
- Gather around a whiteboard and walk through each step in your process – use process charting tools or stick figures and arrows
- If possible, quantify the problem size. Graph your data if possible, visualizations are really helpful

2: Conduct a root cause analysis

- To figure out the root cause of the efficiencies. You can ask questions like:
  - Where do we suffer from communication breakdowns?
  - Where do we see long delays without activity?
  - What information are we needing to collaborate more effectively/smoothly?
- The 5 whys is a helpful tool for conducting a thorough root cause analysis.

3: Devise countermeasures to address root causes

- Countermeasures are your ideas for tackling the situation; the changes to be made to your processes that will move the organization closer to ideal by addressing root causes. Countermeasures should aim to:
  - Specify the intended outcome and the plan for achieving it.
  - Create clear, direct connections between people responsible for steps in the process.
  - Reduce or eliminate loops, workarounds, and delays.

4: Define your target state

- Once you've selected your countermeasures, you are able to clearly define your target state. In the A3 process, you communicate our target state through a process map. Be sure to note where the changes in the process are occurring so they can be observed.

5: Develop a plan for implementation

- Now that you've defined your target state, you can develop a plan for how to achieve it. Implementation plans should include:
  - A task list to get the countermeasures in place
  - Who is responsible for what
  - Due dates for any time-sensitive work items
- Most teams choose to document their implementation plan in their A3.

6: Develop a follow-up plan with predicted outcomes

- A follow-up plan allows Lean teams to check their work; it allows them to verify whether they actually understood the current condition well enough to improve it. A follow-up plan is a critical step in process improvement because it can help teams make sure the:
  - implementation plan was executed
  - target condition was realized
  - expected results were achieved

- These first six steps are captured in the A3 report. Most teams use a template for their A3.

A3 Report: Project name

AMA | STEPSforward

Project mission statement
What is the team trying to accomplish?

Background
- Problem background
- Why the problem needed to be fixed
- Importance of identifying solution

Original state/problem statement
- Use a diagram if possible
- Show where the problems exist with Kaizen bursts, i.e. graphic indicators of opportunities for improvement
- Extent of the problem (e.g., metrics or measures of success that are below target)



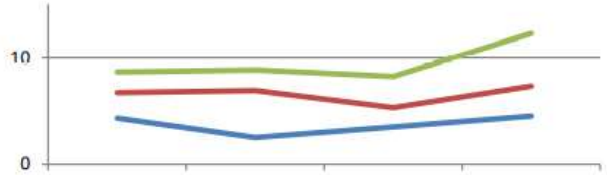
Problem analysis
- Why does the problem exist?
- Does asking "why?" five times help identify the root cause?
- What influences caused the problem?

Team
Executive sponsor:
Champion:
Team lead(s):
Project team: List names and departments

Solution
- Describe recommendations of team
- Show diagram or map of new process
- Measurable targets to achieve within determined timeframes

Implementation plan
- Use a diagram if possible
- Who is responsible for which tasks?
- What resources are required?
- What targets have been identified? Timeline for achieving?
- How regularly will the improvement team connect while the change is underway?

Graph results
Show improvement over time



Sustain
Implementing a project doesn't guarantee long-term success. How does the team plan to continue to make the improvement part of daily practice, long after the "project" as ended? Determine metrics to track, feedback loops for staff, and maintain regular A3 updates to share with the team and supporting leadership.

Source: AMA. Practice transformation series: starting lean healthcare. 2015.

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A3 Project Title

Project Lead:
Project Champion(s):
Date Updated:

Project Team:

UCLA Health

1) Problem Statement: (description of the problem and its effect)

2) Current State: (depiction of the current state, its processes, and problem(s)

Best Practices/Literature Search:

3) Goal: (how will we know the project is successful; standard/basis for comparison)

4) Root Cause Analysis: (investigation depicting the problems' root causes)

5) Solutions: (action plan and findings of tested solutions)

Root Cause	Tested Solution	Responsible	Due	Finding

6) Check: (Summary of the solutions' results, overall goal success, and any supporting metrics)

Goal & Metrics	Baseline	Target	Current
Goal			
Supporting Metric			
Supporting Metric			

7) Act: (Action taken as a result of the Check, and the plan to sustain results)

*A3 is a UCLA Operating System 11x17 template used to document and communicate complex problem-solving using the Plan Do Check Act (PDCA) method: Steps 1-4 (Plan), Step 5 (Do), Step 6 (Check), Step 7(Act)

A3 PROBLEM SOLVING TOOL:

Contact:

Date:

BACKGROUND / BUSINESS CASE

- What issue or problem do we need to solve?
- Why is this issue important to solve now?
- What benefits do we anticipate from solving the problem (e.g., quality, timeliness, cost, customer/employee satisfaction)?

STAKEHOLDERS

- Who are internal and external customers?
- Who are team members that will complete the A3 Problem Solving Tool?

CURRENT CONDITION

- What do we know? What customer, process, program data/measures do we have on the problem (location, patterns, trends, frequency, factors)? Answer questions like: ***What*** errors are occurring? ***Who*** is making the errors? ***Where*** are the errors occurring? ***When*** are the errors occurring? **How** are the errors occurring?
- What don't we know and need to find out? We may need to develop a Data Collection Plan that includes: The information/data we need to collect, who will collect the data, data sources, who will prepare the visuals (bar chart, trend, pie chart), when and who will be sent the data.
- What is the Problem Statement? What specific performance measure needs to improve? We need to understand the scope and nature of the problem before we can create a problem statement. More analysis may be needed if the team cannot write a problem statement.
  - Example: Reduce/Increase the number/percent of <?> from <current level> to <desired level> by <date>.

ANALYSIS/ROOT CAUSES

- What are root causes? Why are the errors occurring?
  - If the root cause is not obvious, use a root cause analysis tool. Use the simplest tool to show cause-and-effect down to the root cause(s). The root cause should be specific – not vague like "poor communication".
  - Tools: 5 Whys, Fishbone diagram, or Affinity and Relations diagrams
- Does our data verify the root causes? – a team may need to collect additional data to verify the root cause(s)

SOLUTIONS

What solutions will solve the root causes? (Tools: Brainstorming and Affinity Diagram)

- What solutions are best and we should recommend?
  - Tool for a few primary options: Impact/Difficulty Matrix
  - Tool for many options: Criteria Decision Matrix
  - Consider including an evaluation of the status quo (no change) option
- What impacts (positive and negative) may result from implementing the solutions? (Tool: Impact Wheel, FMEA)
- How will we mitigate or resolve negative impacts?
- What communication or stakeholder engagement is needed? (Tool: Communication Plan)
- What training is needed?

ACTION ITEMS

Task	Owner	Proposed Date	Actual Date

- What tasks or actions do we need to take? Who will be responsible for the task? When should the task be completed? (Tools: Action Plan, Gantt Chart)
- What support and resources are needed for each task?

METRICS/FOLLOW-UP

- What metrics will we use to track progress and performance? How will we validate results?
- How and when will we check progress and performance (e.g., daily, weekly, 30, 60, 90-days)?
- What processes will we use to enable, assure, and sustain success?
- How will we communicate results and share what we learn with others?

A3 TITLE: **Owner:** **Sponsor/Manager:** **Version #:** **DATE:**

Problem Statement (Describe the Problem)

Historical Trend/Background (Current State of the Situation)

Root Cause Analysis

Implementation Plan/Countermeasures/Outcome

WHAT will be done	By WHOM	By WHEN	What was the OUTCOME

Target/Goal(s)

Summary/Wrap-Up/Next Steps

7: Get everyone on board

- The goal for any systemic improvement is that it improves every part of the system. This is why it's vital to include everyone who might be affected by the implementation or the target state in the conversation before changes are made.

7: Get everyone on board

- Building consensus throughout the process is usually the most effective approach, which is why many teams choose to include this at each critical turning point in the A3 process. Depending on the scope of the work, it might also be important to inform executives and other stakeholders who might be impacted by the work.

8: Implement!

- Now it’s time for implementation. Follow the implementation as discussed, observing opportunities for improvement along the way.

9: Evaluate Results

- In far too many situations, the A3 process ends with implementation. It's critical to measure the actual results and compare them to your predictions in order to learn.
- If your actual results vary greatly from what was expected, do research to figure out why. Alter the process as necessary, and repeat implementation and follow-up until the goal is met.